

Bath Planner: the AI Bathroom Designer & Planner

THE PROBLEM

- **Four constraints, no single tool**
Room dimensions, code clearances, product compatibility, and budget all have to line up at once.
- **Showrooms sell options, not plans**
A catalog shows what exists, not what actually fits an exact 5'×8' room.
- **Mistakes surface too late**
A fixture picked for looks can turn out not to clear the door swing, or blow the budget, only after it's already ordered.

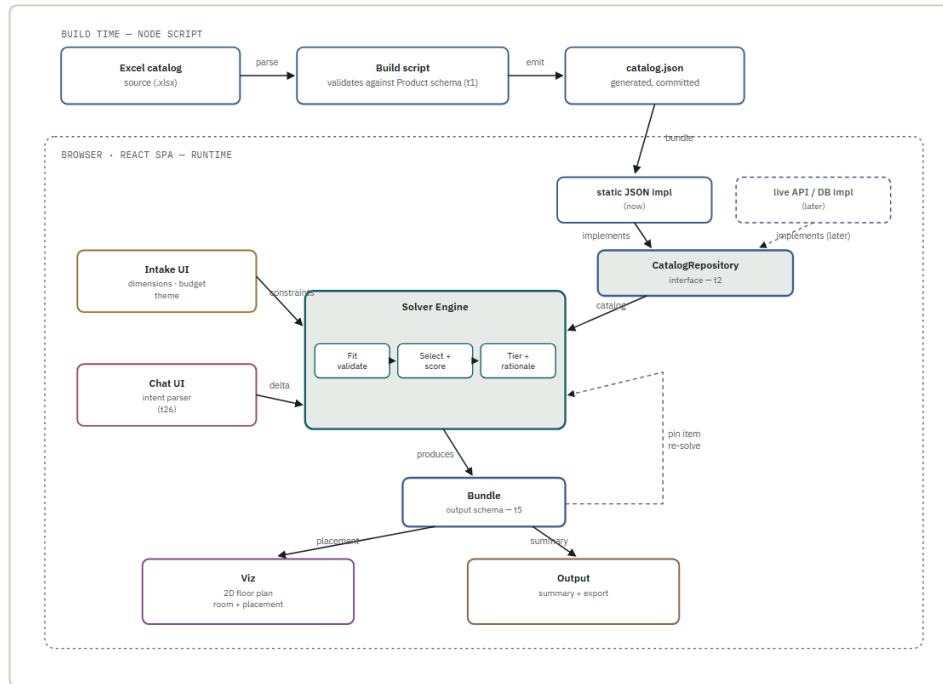
THE SOLUTION

- **A real constraint solver**
Takes room dimensions, a budget, and a style, and returns bundles already checked against real clearances.
- **Three ready-made tiers**
Value, Balanced, and Premium, each fully priced with a plain-language reason for every pick.
- **Refine it by talking to it**
“Swap the vanity for something cheaper” re-solves live and explains what changed and why.

Live: kohlerbathplanner.vercel.app
github.com/Alok49225/kohlerbathplanner

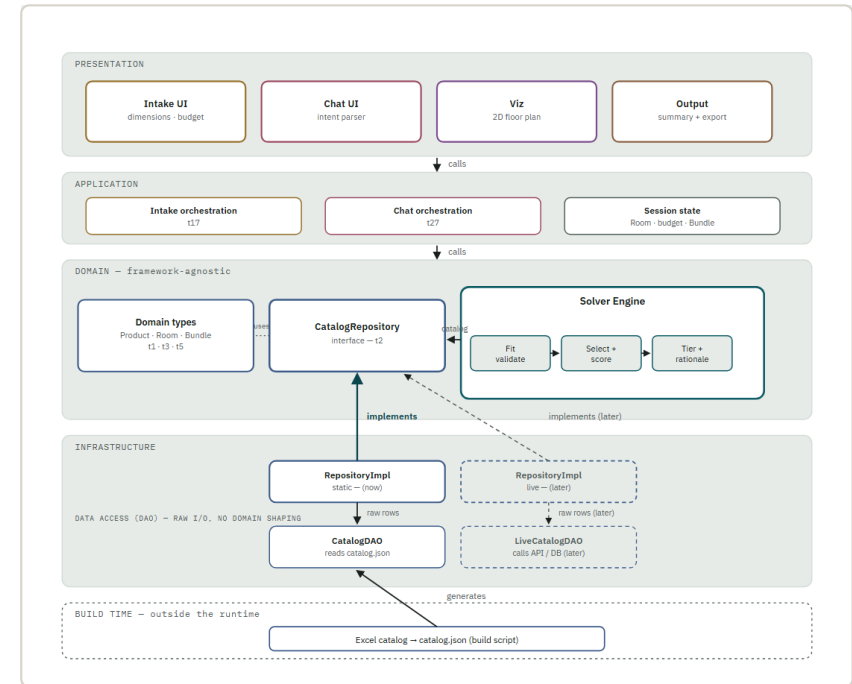
How the System Works

DATA FLOW



Excel → catalog.json at build time; Intake/Chat feed the Solver Engine, which produces a Bundle for Viz + Output.

LAYERS & DEPENDENCY DIRECTION



Every arrow points down except one: Infrastructure implements an interface Domain owns.

BUILT AI-ASSISTED, SPEC-FIRST

All 50 build tasks (the original 35, plus 15 more for automatic fixture placement and the product picker) were proposed as a concrete spec, approved, then implemented and tested by Claude before the next began. Every post-launch bug was reproduced and root-caused live in the running app — never guessed. See the Prompts Documentation for the full record.

THE SOLVER PIPELINE

- 1 Auto fixture placement
- 2 Compatibility rules
- 3 Fit & clearance validation
- 4 Budget-constrained optimization
- 5 Multi-objective scoring
- 6 3 tiers, price non-decreasing

Key Features

SHIPPED TODAY

- **Automatic fixture placement**
Auto-places toilet, vanity, and shower with a deterministic search; a fixed fallback ladder drops vanity, then shower, rather than just failing outright.
- **Click-to-place plumbing**
Set rough-ins by clicking the floor plan; wall and clearance direction are inferred automatically.
- **3 auto-priced tiers**
Value, Balanced, and Premium bundles, price guaranteed non-decreasing across tiers.
- **Conversational refinement**
“Swap the vanity for something cheaper” re-solves live and explains the change.
- **Product picker**
Browse every eligible option for a category, sorted by price, and lock one in with a click.
- **Theme-matched flooring**
The floor plan's own color and material name follow your selected style — matte white tile, marble, or wood.
- **Explainable rationale**
Every line item states why it was chosen, not just what it costs.
- **Water-efficiency scoring**
Every bundle carries a sustainability score, surfaced directly in the summary.

ON THE ROADMAP — NOT YET BUILT

AI-rendered room visualization

An AI image model generating a photo-realistic view of the solved bundle inside the actual room, clearly labeled as illustrative rather than a literal product photo — the “AI concept render” already scoped as a Could-have in the original blueprint, not a new idea invented for this deck.

Technology & Results

React + TS + Vite

Fully static, zero backend

384 / 384

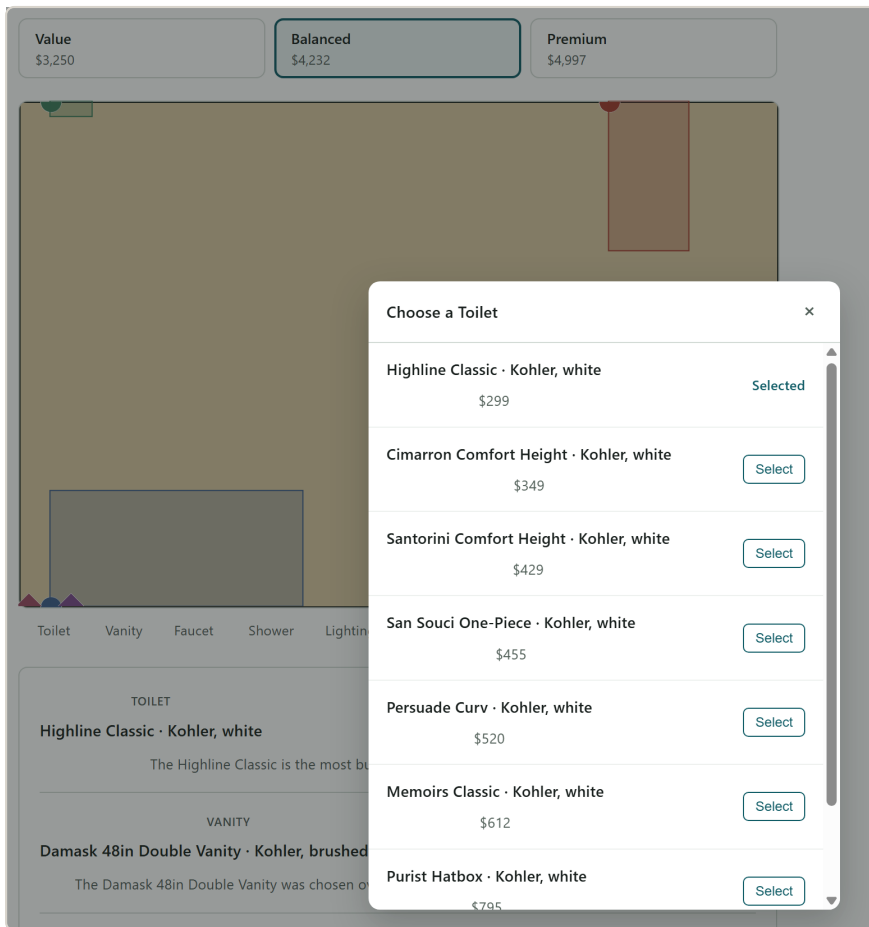
automated tests passing

50 / 50

build tasks complete

Live

kohlerbathplanner.vercel.app

11real bugs found & fixed
post-launch**THE WORKING MODEL — LIVE ON KOHLERBATHPLANNER.VERCEL.APP***Product picker — every eligible option, sorted by price, live**Auto-placed fixtures, theme-matched flooring, explainable pricing*